SAT Unit 4.1
Volume and Area

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Excellence in SAT Prep

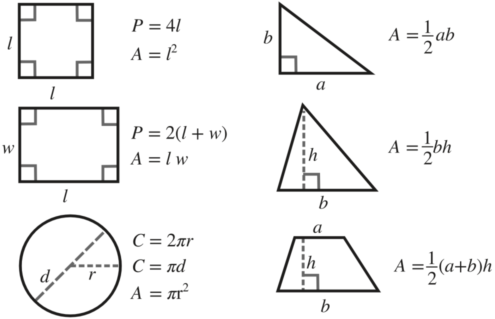
Overview

1. Volume and Area

2. Practice

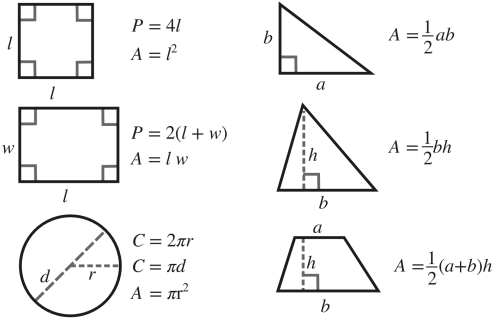
Perimeter Formulas

- Square: $P = 4s$
- Rectangle: $P = 2(l + w)$
- Triangle: $P = a + b + c$
- Circle (Circumference): $P = 2\pi r$ or $P = \pi d$



Area Formulas

- Square: $A = s^2$
- Rectangle: $A = lw$
- Triangle: $A = \frac{1}{2}bh$
- Parallelogram: $A = bh$
- Trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$
- Circle: $A = \pi r^2$



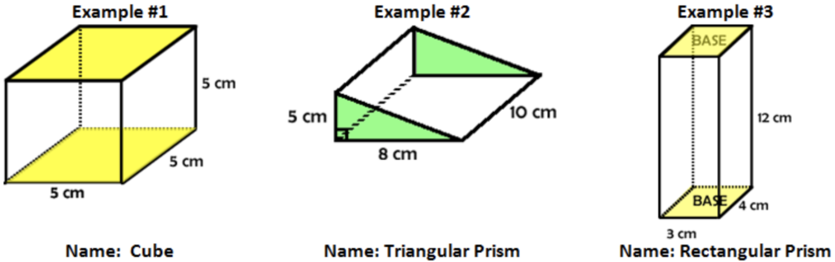
Prism Volume

Definition: A prism is a three-dimensional shape with two parallel and congruent bases. The volume of a prism is calculated using the formula:

$$V_{\text{PRISM}} = Bh$$

where:

- B is the **Area of the Base**.
- h is the **Height of the Prism** (one of its lateral sides).



Surface Area of a Cube

Question: A cube has a volume of 42,875 cubic inches. What is the surface area, in square inches, of the cube?

Solution

Solution:

- Let s be the side length of the cube.
- Given:

$$s^3 = 42,875$$

- Solve for s:

$$s = \sqrt[3]{42,875} \approx 35 \text{ inches}$$

- Surface area of the cube:

$$\text{Surface Area} = 6s^2 = 6 \times (35)^2 = 6 \times 1225 = 7350 \text{ in}^2$$

Answer: 7350 square inches.

Cylinder Volume Calculation

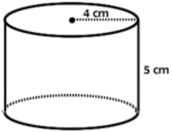
Definition: A cylinder is a three-dimensional shape with two parallel, congruent circular bases. The volume of a cylinder is calculated using the formula:

$$V_{\text{CYLINDER}} = Bh$$
$$V_{\text{CYLINDER}} = \pi r^2 h$$

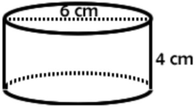
where:

- B is the **Area of the Base** (which is a circle, so $B = \pi r^2$).
- h is the **Height of the Cylinder**, the perpendicular distance between the two circular bases.

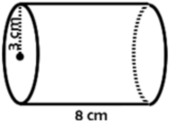
Example #1



Example #2



Example #3



Cylinder Problem

Question: A cylinder has a volume of 980π cubic feet and a height of 5 feet. What is the base diameter, in feet, of the cylinder?

Solution

Solution:

- Volume of a cylinder:

$$V = \pi r^2 h$$

- Given:

$$980\pi = \pi r^2 \times 5$$

- Simplify:

$$r^2 = \frac{980}{5}$$

$$r^2 = 196$$

$$r = 14 \text{ feet}$$

- Diameter:

$$d = 2r = 2 \times 14 = 28 \text{ feet}$$

Answer: 28 feet

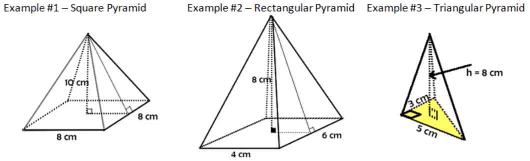
Pyramid Volume Calculation

Definition: A pyramid is a three-dimensional shape with a polygonal base and triangular faces that converge at a single point (the apex). The volume of a pyramid is given by the formula:

$$V_{\text{PYRAMID}} = \frac{1}{3}Bh$$

where:

- B is the **Area of the Base**.
- h is the Height of the Pyramid, the perpendicular distance from the base to the apex.
- The volume of a pyramid is always **one-third** of the volume of a prism with the same base and height.



Volume of a Pyramid

Question: A right square pyramid has a height of 33 centimeters (cm) and a square base with a side length of 14 cm. What is the volume, in cm^3, of this pyramid?

Solution

Solution:

- The formula for the volume of a right square pyramid is:

$$V = \frac{1}{3} \times \text{Base Area} \times \text{Height}$$

- Base area:

$$\text{Base Area} = 14 \times 14 = 196 \text{ cm}^2$$

- Therefore:

$$V = \frac{1}{3} \times 196 \times 33$$

$$V = \frac{1}{3} \times 6468$$

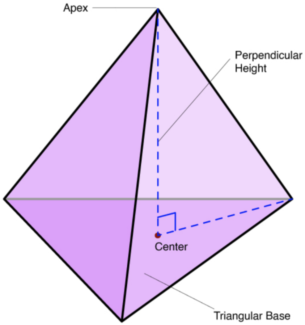
$$V = 2156 \text{ cm}^3$$

Answer: 2156 cm^3

Right Triangular Pyramid

What is a Right Triangular Pyramid?

- A three-dimensional solid with a triangular base.
- The apex (top point) is directly above the centroid of the triangular base.
- Has 4 faces: 1 triangular base and 3 triangular lateral faces.
- All lateral edges meet at the apex.



Right Triangular Pyramid

Right Triangular Pyramid

Volume Formula:

$$V = \frac{1}{3} \times \text{Base Area} \times \text{Height}$$

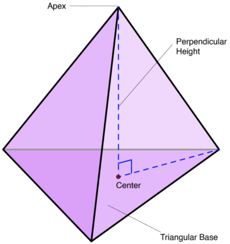
Where:

- Height = perpendicular distance from the apex to the base

Surface Area Formula:

$$\text{Surface Area} = \text{Base Area} + \text{Sum of lateral face areas}$$

Each lateral face is a triangle: $\frac{1}{2} \times \text{base} \times \text{slant height}$



Right Triangular Pyramid

Density

Definition: Density is a measure of how much mass is contained in a given volume.

Formula

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

where:

- Mass is typically measured in grams (g) or kilograms (kg).
- Volume is typically measured in cubic centimeters (cm^3) or cubic meters (m^3).

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Example:

- A rock has a mass of 200 grams and a volume of 50 cm^3.
- Calculate the density:

$$\text{Density} = \frac{200 \text{ g}}{50 \text{ cm}^3} = 4 \text{ g/cm}^3$$

Density Problem

Question: The mass of a foam block is 8 kilograms. The volume of the block is 20 cubic meters. What is the density, in kilograms per cubic meter, of the block?

A. 0.4

B. 2.5

C. 12

D. 28

Solution

Solution: Density is given by:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{8\ \text{kg}}{20\ \text{m}^3} = 0.4\ \text{kg/m}^3.$$

Answer: A.

Ratio of Length, Area, and Volume

Concept: When a geometric shape is scaled by a factor of k, the following relationships hold:

- **Length Ratio:** If the scale factor is k, then all linear measurements (length, width, height, radius, etc.) change by a factor of k.

$$\text{New Length} = k \times \text{Original Length}$$

- **Area Ratio:** The area of a shape is proportional to the square of the scale factor.

$$\text{New Area} = k^2 \times \text{Original Area}$$

- **Volume Ratio:** The volume of a three-dimensional shape is proportional to the cube of the scale factor.

$$\text{New Volume} = k^3 \times \text{Original Volume}$$

Question: Ratio of Length, Area, and Volume

Suppose a cube has an original edge length of 2 cm, and we scale it up by a factor of $k = 3$. How do the length, area, and volume change under this transformation?

Options:

- A Length scales by k, area scales by k^2, volume scales by k^3.
- B Length scales by k^2, area scales by k^3, volume scales by k^4.
- C Length scales by k, area scales by k^3, volume scales by k^2.
- D Length scales by k^3, area scales by k^2, volume scales by k.

Answer Explanation: Ratio of Length, Area, and Volume

Correct Answer: $\mathbf{k, k^2, k^3}$

Solution: When a shape is scaled by a factor of k:

- **Length:** Scales by k.

$$\text{New Length} = k \times \text{Original Length}$$

- **Area:** Scales by k^2.

$$\text{New Area} = k^2 \times \text{Original Area}$$

- **Volume:** Scales by k^3.

$$\text{New Volume} = k^3 \times \text{Original Volume}$$

Final Answer: $\boxed{k, k^2, k^3}$.

Question: Finding the Volume of a Cube

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Options:

A. 18

B. 27

C. 36

D. 81

Answer Explanation: Finding the Volume of a Cube

Correct Answer: B

Solution: A cube has 6 identical square faces. If the total surface area is given as:

$$6s^2 = 54$$

Solve for s:

$$s^2 = \frac{54}{6} = 9$$

$$s = \sqrt{9} = 3$$

Now, calculate the volume:

$$V = s^3 = 3^3 = 27$$

Question: Area of a Scaled Poster

A rectangular poster has an area of **360** square inches. A copy of the poster is made in which the **length** and **width** of the original poster are each increased by **20%**.

What is the area of the copy, in square inches?

Answer Explanation: Area of a Scaled Poster

Solution:

Step 1: Define Scaling Factor Since both length and width increase by 20%, the new dimensions are scaled by:

$$k = 1.2$$

Step 2: Apply Area Scaling Formula Since **area scales by k^2**:

$$\text{New Area} = 360 \times (1.2)^2$$

$$= 360 \times 1.44 = 518.4$$

Final Answer: $\boxed{518.4}$ square inches.

Key Takeaways

- Memorizing perimeter, area, and volume formulas is crucial.
- Understanding the ratio between the length, Area, and the Volume.
- Recognizing shapes and their properties in SAT problems.

Practice Makes Perfect!

Cube Problem

Question: The surface area, in square inches, of a cube can be represented by the expression $24a^2$, where a is a positive constant. Which expression represents the volume, in cubic inches, of the cube?

A) $8a^3$

B) $16a^3$

C) $64a^6$

D) $216a^6$

Solution

Solution:

- Let s be the side length of the cube.
- The surface area of the cube is $6s^2 = 24a^2$.
- Solve for s^2:

$$s^2 = \frac{24a^2}{6} = 4a^2$$

- Then:

$$s = 2a$$

- The volume of the cube:

$$V = s^3 = (2a)^3 = 8a^3$$

Answer: $8a^3$ (Choice A)

Volume of a Cylinder

Question: Given the function $f(x) = \pi x^3 + \pi x^2$, where f represents the volume (in cubic feet) of a right circular cylinder with radius x feet, which expression represents the height (in feet) of the cylinder?

A) $x + 1$
B) πx^3
C) x
D) $x^3 + 1$

Solution

Solution:

- The volume of a right circular cylinder is given by:

$$V = \pi r^2 h$$

- Comparing with $f(x) = \pi x^3 + \pi x^2$, notice that:

$$f(x) = \pi x^2(x + 1)$$

- Therefore, $r = x$ and $h = x + 1$.

Answer: $x + 1$ (Choice A)

Rectangle Perimeter Problem

Question: Identical rectangles X and Y each have an area of $\frac{49}{5}$ square meters and a perimeter of P meters. If one of the shorter sides of rectangle X is glued to one of the shorter sides of rectangle Y, the resulting rectangle has a perimeter of $\frac{14}{9}P$ meters. What is the value of P?

Solution

Solution:

- Let l and w be the length and width of each rectangle.
- Then:

$$lw = \frac{49}{5} \quad \text{and} \quad 2(l + w) = P$$

- When glued along the shorter side:

$$\text{New perimeter} = 2(l + 2w)$$

- Given:

$$2(l + 2w) = \frac{14}{9} P$$

- But $P = 2(l + w)$ so:

$$2(l + 2w) = \frac{14}{9} \cdot 2(l + w)$$

- Dividing both sides by 2:

$$l + 2w = \frac{14}{9}(l + w)$$

- Solve:

$$9(l + 2w) = 14(l + w)$$
$$9l + 18w = 14l + 14w$$
$$4w = 5l$$

Solution

- From $lw = \frac{49}{5}$:

$$l\left(\frac{5l}{4}\right) = \frac{49}{5}$$

$$\frac{5l^2}{4} = \frac{49}{5}$$

$$l^2 = \frac{49}{5} \cdot \frac{4}{5} = \frac{196}{25}$$

$$l = \frac{14}{5}$$

- Then:

$$w = \frac{5l}{4} = \frac{5 \cdot \frac{14}{5}}{4} = \frac{14}{4} = 3.5$$

- Hence:

$$P = 2(l+w) = 2\left(\frac{14}{5} + 3.5\right) = 2\left(\frac{14}{5} + \frac{7}{2}\right)$$

$$= 2\left(\frac{28}{10} + \frac{35}{10}\right) = 2\left(\frac{63}{10}\right) = \frac{126}{10} = 12.6$$

Answer: $P = 12.6$ meters.

Thank You!

We appreciate your time and focus.

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